

Jingxuan (Jasper) Wu

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Research interests

Controllable generative AI; diffusion and flow-based models; reinforcement learning and post-training; long-context memory; agentic systems.

Education

The University of North Carolina at Chapel Hill Chapel Hill, NC
M.S. in Statistics, Analytics, and Data Science Aug. 2025 – Dec. 2026 (expected)

The Chinese University of Hong Kong, Shenzhen Shenzhen, China
B.S. (Hons) in Data Science and Big Data Technology Sep. 2021 – May 2025

Publications

Conference papers

Letting Trajectories Spread: Quality-Preserving Control for Diverse Flow Matching.

Jingxuan Wu, Zhenglin Wan, Xingrui Yu, Yuzhe Yang, Bo An, Ivor Tsang, and Yang You. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

Adversarial Dual On-Policy Distillation from Expressive Teacher.

Zhenglin Wan, **Jingxuan Wu**, Xingrui Yu, Chubin Zhang, Mingcong Lei, Bo An, Ivor W. Tsang, and Yang You. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

Denoising Time Matters: Diverse Generation in Diffusion Language Models.

Jingxuan Wu, Zhenglin Wan, Yuzhe Yang, Yiqiao Huang, Chubin Zhang, Xingrui Yu, Ivor Tsang, and Yang You. *40th Conference on Neural Information Processing Systems (NeurIPS)*, 2026.

Calibration Is Not Control: Why LLM-Agent Oversight Needs Intervention.

Chubin Zhang, Zhenglin Wan, Xingrui Yu, **Jingxuan Wu**, Qi Wen, Pengfei Zhou, Wangbo Zhao, and Ivor Tsang. *40th Conference on Neural Information Processing Systems (NeurIPS)*, 2026.

Journal paper

Flight Delay Propagation Modeling: Data, Methods, and Future Opportunities.

Chi Li, Jianfeng Mao, Lingyi Li, **Jingxuan Wu**, Lianmin Zhang, Jianyu Zhu, and Zibin Pan. *Transportation Research Part E: Logistics and Transportation Review*, 185:103525, 2024.

Preprints and technical reports

MemFold: Learning Compact Soft Memory for Long-Context Personalization via On-Policy Optimization.

Jingxuan Wu, Yuzhe Yang, Yiqiao Huang, Chengzhi Liu, Qingni Wang, Chengxuan Qian, Shutong Wu, Jiawei Zhang, and Xin Eric Wang. Under review at ICLR 2027.

LLM-Powered Predictive Decision-Making for Sustainable Data Center Operations.

Hanzhao Wang*, **Jingxuan Wu***, Yumeng Li, Yu Pan, and Guanting Chen. arXiv preprint, 2026. *Equal contribution.

CaveAgent: Transforming LLMs into Stateful Runtime Operators.

Maohao Ran, Zhenglin Wan, Cooper Lin, Yanting Zhang, Hongyu Xin, Hongwei Fan, Yibo Xu, Beier Luo, Yaxin Zhou, Wangbo Zhao, Lijie Yang, Lang Feng, Fuchao Yang, **Jingxuan Wu**, Yiqiao Huang, Chendong Ma, Dailing Jiang, Jianbo Deng, Sihui Han, Bo An, Yike Guo, and Jun Song. Technical report, 2026.

Research experience

University of Wisconsin–Madison

May 2026 – Present

Advisor: Prof. Jiawei Zhang

Madison, WI

Research on the **generalization of LLM agents** across search, coding, and memory-augmented question answering, with an emphasis on transferable memory and decision policies that remain effective on unseen tasks and out-of-distribution datasets.

Representative work: [MemFold](#); generalizable Coding & Web-Search Agent (ongoing).

A*STAR Centre for Frontier AI Research

May 2025 – Present

Advisors: Xingrui Yu, Ivor Tsang, Bo An

Singapore and Remote

Research on **reinforcement learning and post-training for generative models**, including diversity-aware sampling as a mechanism for broader exploration and more informative training data, as well as on-policy distillation for efficient embodied control and intervention-based oversight for LLM agents.

Representative papers: [TAPS](#); [OSCAR](#); [FA-OPD](#); [Calibration Is Not Control](#).

The University of North Carolina at Chapel Hill

Jan. 2024 – Present

Advisors: Guanting Chen, Xiaocheng Li, Wenjia Ba

Chapel Hill, NC and Remote

Research at the intersection of **operations research and artificial intelligence**: combining LLM-based workload prediction with online learning for data-center resource allocation, and integrating diffusion-based product generation with utility modeling to learn consumer preferences from limited human feedback.

Representative work: [LLM-DC](#); [SafeScreen](#).

Projects

CaveAgent

Oct. 2025 – Jan. 2026

Open-source contributor | Python, LLM agents, AST, Asyncio

- Contributed persistent Python runtime modules supporting object injection and retrieval across stateful, multi-turn LLM interactions.
- Implemented AST-based rules to validate and sanitize LLM-generated code, and developed discovery/registry mechanisms for modular Agent Skills.
- Participated in benchmarking across OpenAI, Claude, and Gemini backends to evaluate framework robustness.